Gauge coupling constants from tetrahedral combinatorics

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Abstract

We derive the strong coupling constant $a_s(m_Z)$, the weak mixing angle $\sin^2 th_W(m_Z)$, and the inverse fine structure constant $a_{EM}^{-1}(m_Z)$ from the combinatorial geometry of the complete graph K_4 with zero free parameters. The Standard Model gauge group SU(3) x SU(2) x U(1) emerges from the face, edge, and vertex structure of K_4. Coupling ratios follow from boundary element counting, and the overall normalization is fixed by the Regge deficit angle on the K_4 two-skeleton. We obtain $a_s = 129/(112\pi^2) = 0.1167$ (measured 0.1180; deviation 1.1%), $\sin^2 th_W = 3/13 = 0.2308$ (measured 0.23121; deviation 0.19%), and $a_{EM}^{-1} = 13\pi^2 = 128.3$ (measured 127.951; deviation 0.28%). Additional predictions include the neutron-proton mass difference $(m_n - m_p)/m_p = 1/(72\pi^2)$ (2.1%), the matter sector coupling $a_{matter}/a_{dist} = 1/d^3 = 1/27$, and the lepton generation count $N_{gen} = 3$ from $A_4/V_4 = Z_3$ with Koide ratio $Q = 2/3$ as the Z_3 character saturation.

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I. INTRODUCTION

The Standard Model of particle physics contains 19 free parameters whose values are determined by experiment rather than by the theory itself [1]. Among these, the gauge coupling constants set the strengths of the strong, weak, and electromagnetic interactions. Despite decades of effort in grand unification [2,3] and string theory [4], no framework has derived these constants from geometry alone with zero adjustable parameters.

We present such a derivation. The complete graph K_4 -- four vertices connected by all six edges, forming a tetrahedron -- serves as the sole geometric input. The Standard Model gauge group, coupling ratios, overall normalization, generation count, and matter sector structure all follow from the combinatorial properties of K_4 and the Regge calculus on its two-skeleton.

II. THE K_4 FRAMEWORK

The complete graph K_4 has $V = 4$ vertices, $E = 6$ edges, $F = 4$ triangular faces, and Euler characteristic chi = 2. Its symmetry group is S_4 (order 24), with alternating subgroup A_4 (order 12) and Klein four-group V_4 (order 4). We assign gauge content by boundary structure: faces (3 boundary edges) carry SU(3), edges (2 boundary vertices) carry SU(2), and vertices (1 self-boundary) carry U(1).

III. COUPLING DERIVATION

A. Ratios from boundary counting

The gauge coupling g_p^2 on a p-simplex is proportional to its boundary count: $g_p^2 \sim (p+1)$. This gives $g_3^2 : g_2^2 : g_1^2 = 3:2:1$. With SU(5) normalization:

$$\sin^2 th_W = (3/5)/(3/5+2) = 3/13 = 0.23077$$

B. Normalization from the Regge deficit angle

On the K_4 two-skeleton, S_4 symmetry forces equilateral faces with interior angle pi/3 at each vertex. Each vertex belongs to d = 3 faces. The Regge deficit angle at each vertex is:

$$\textbf{epsilon = 2pi - } d\textbf{(pi/3) = 2pi - pi = pi}$$

This is exact and unique to d = 3. Total curvature: 4pi = 2pi chi. The gauge coupling equals boundary count divided by the curvature quantum:

$$g_p^{\,2} = (p+1)/\text{pi}$$

IV. THE STRONG COUPLING CONSTANT

From $g_3^{\,2}$ = 3/pi: a_{dist} = 3/(4pi^2) = (2pi chi)^2/|A_4|. The support-completeness theorem gives S_0 Delta = E^2+E+1 = 43, decomposing uniquely as 36 (edge-edge) + 6 (identity-cross) + 1 (identity-self).

$$a_s(m_Z) = a_{dist} \text{ x } S_0/V = 129/(112\text{pi}^2) = 0.1167 \text{ (measured 0.1180, 1.1\%)}$$

V. ELECTROWEAK PREDICTIONS

sin^{2}th$_W$ = **3/13 = 0.23077** (measured 0.23121, 0.19%). The SU(2) coupling receives edge support factor 2/d = 2/3: a_2 = 1/(3pi^2). Combined:

$$a_{EM}^{\,-1} = 1/(a_2 \sin^2\text{th}) = 13\text{pi}^2 = 128.3 \text{ (measured 127.951, 0.28\%)}$$

The integer 13 = V+E+F-1 counts the proper boundary simplices of K_4.

VI. THREE-POLE SECTOR STRUCTURE

The topological selection principle F_{IFS}(chi) = 2/(3pi chi) selects chi = 2 uniquely via orientability, positive curvature, and finite contraction. The screening constant epsilon* = d F_{IFS}/|A_4| = 1/(12pi) satisfies the exact identity:

$$\textit{epsilon* x } |A_4| = 1/\text{pi (total screening = inverse deficit)}$$

The matter sector coupling a_{matter} = a_2/|A_4| = 1/(36pi^2) is suppressed relative to the geometry coupling by:

$$a_{matter}/a_{dist} = 1/d^3 = 1/27$$

One factor of 1/d per spatial dimension. The SU(2) coupling is the cross-sector mediator: a_2 = d x $F_{IFS}^{\,2}$.

Neutron-proton mass difference:

$$(m_n - m_p)/m_p = a_{matter} \text{ x } d/E = 1/(72\text{pi}^2) = 0.001407 \text{ (measured 0.001378, 2.1\%)}$$

VII. GENERATION STRUCTURE AND THE KOIDE FORMULA

The quotient A_4/V_4 = Z_3 yields three cosets corresponding to three lepton generations. The Z_3 characters are equally spaced at 2pi/3, and the Koide parametrization of charged-lepton masses is the Z_3 character expansion:

$$\textit{sqrt}(m_l) = M_K(1 + \textit{sqrt}(2)\cos(\text{th} + 2\text{pi } l/3)),\ l = 0,1,2$$

The Koide ratio Q = 2/3 = (d-1)/d saturates the maximum agreement achievable with d = 3 faces per vertex (odd-d frustration theorem). The color enhancement factor $2^{C(p+1,2)}$ for quark masses follows from the number of internal edges in each generation's simplex.

VIII. INCIDENCE-HODGE BRIDGE

The complex structure J = (1/4)[BB^T, $*_E$] satisfies J^2 = -I on the K_4 edge space. The composite JG (complex structure times Z_3 generator) has block-off-diagonal form, with the chiral transfer block:

$$D = [[0,0,-1],[1,0,0],[0,-1,0]] \text{ with eigenvalues } \{1, \omega, \omega^2\}$$

The standard generation generator P = QDQ^{-1} (where Q = diag(1,1,-1)) recovers the Z_3 character basis z_0 = (1,1,1), z_1 = (1,ω^2,ω), z_2 = (1,ω,ω^2). The complete chain:

$$K_4 \text{ incidence} \dashrightarrow \text{edge Hodge star} \dashrightarrow J \dashrightarrow D \dashrightarrow QDQ^{-1} = P$$

IX. DISCUSSION

Table I. Predictions from K_4 combinatorics. No free parameters.

Observable	Formula	Prediction	Measured	Error
a_s(m_Z)	129/(112pi^2)	0.1167	0.1180(9)	1.1%
sin^2 th_W	3/13	0.2308	0.23121(4)	0.19%
a_EM^-1	13pi^2	128.3	127.951(9)	0.28%
(m_n-m_p)/m_p	1/(72pi^2)	0.001407	0.001378	2.1%
a_matter/a_dist	1/27	0.03704	—	pred.
N_gen	|A4/V4|	3	3	exact
Q_Koide	(d-1)/d	2/3	0.6667	0.001%
n_s	185/192	0.9640	0.9649(42)	0.09%

Uniqueness of K_4. -- The vertex deficit angle equals pi only for d = 3. The spectral identity $\langle|E_{face}|\rangle = E/V$ holds only for K_3 and K_4. The evaluation saturation theorem shows that the identity self-support (+1) is the precise quantum needed for any complete graph to sustain a distinction.

Falsifiability. -- A determination of $a_s(m_Z)$ deviating from 129/(112pi^2) by more than 3%, or $\sin^2\text{th}_W$ deviating from 3/13 beyond 5 sigma, would falsify this framework.

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